

- | | | | |
|--------|---|----------|----------|
| 1. (a) | $Q = m_s c_s \Delta T + m_s l_v$ $= 0.02 (2000)(110 - 100) + 0.02 (2260000)$ $= 400 + 45200$ $= 45600 \text{ J}$ | 1M+1M | |
| | | 1A | <u>3</u> |
| (b) | $m_m c_m \Delta T_m = Q + m_s c_w \Delta T_w$ $0.2 (3900)(T - 15) = 45600 + 0.02 (4200)(100 - T)$ $T = 76.0 \text{ } ^\circ\text{C}$ | 1M | |
| | | 1A | <u>2</u> |
| (c) | The actual temperature of frothy milk is lower than the value calculated in (b).
Because energy loss by the steam is also gained by the surroundings including the air/the metallic jug that holds the milk. | 1A
1A | <u>2</u> |

2. (a)

$$p_1V_1 = p_2V_2 \text{ (Or } p \propto \frac{1}{V} \text{)}$$

1M

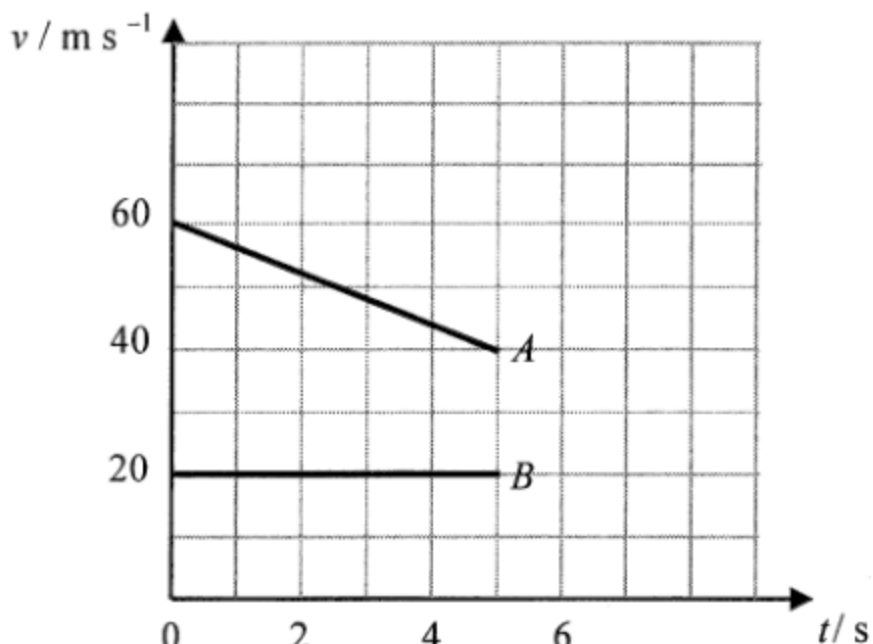
3. (a) (i) Friction f between the tyres and the road.

$$f = \frac{mv^2}{r}$$

$$8000 = \frac{1200 v^2}{45 \text{ m}}$$

$$v = 17.3 \text{ m s}^{-1}$$
1A 3
- (ii) Smaller
For the same f , $v^2 \propto r$; when r decreases v decreases.
1A 2
- (b) (Max) friction / coefficient of friction reduced,
not enough to provide the centripetal force / acceleration required for circular motion.
Or Tracking or allowed speed lowered.
1A 2

Marks

4. (a) (i) $v = u + at$
 $= 60 + (-4)5$
 $= 40 \text{ m s}^{-1}$
1M
1A 2
- (ii)
- 
- 1A
- (iii)
- $$s_A = \left(\frac{60 + 40}{2} \right) (5) = 250 \text{ m}$$
- $$s_B = (20)(5) = 100 \text{ m}$$
- $$x = 250 - 100$$
- $$= 150 \text{ (m)}$$
- [equals to the area between the two graphs]
- (b) (i) $m u_A + m u_B = (m + m)V$
 $40 + 20 = 2V$
 $V = 30 \text{ m s}^{-1}$
1M
1A 2
- (ii)
- $$F = \frac{mV - mu_A}{\Delta t}$$
- $$= (5000) \frac{(30 - 40)}{0.2}$$
- $$= -250000 \text{ N}$$
- Impact force is opposite to the direction of travel of A (to the left / backwards / negative)
1A 3

$$p_1 \left(\frac{4}{3} \pi \times (0.8)^3 \right) = (1.01 \times 10^5) \left(\frac{4}{3} \pi \times (1.0)^3 \right)$$

$$p_1 = 1.97 \times 10^5 \text{ Pa}$$

1A 2

- (b) Volume increases as bubble rises but the speed / k.e. of gas molecules remains unchanged
therefore frequency of collision of molecules on bubble's inner surface decreases, gas pressure decreases.

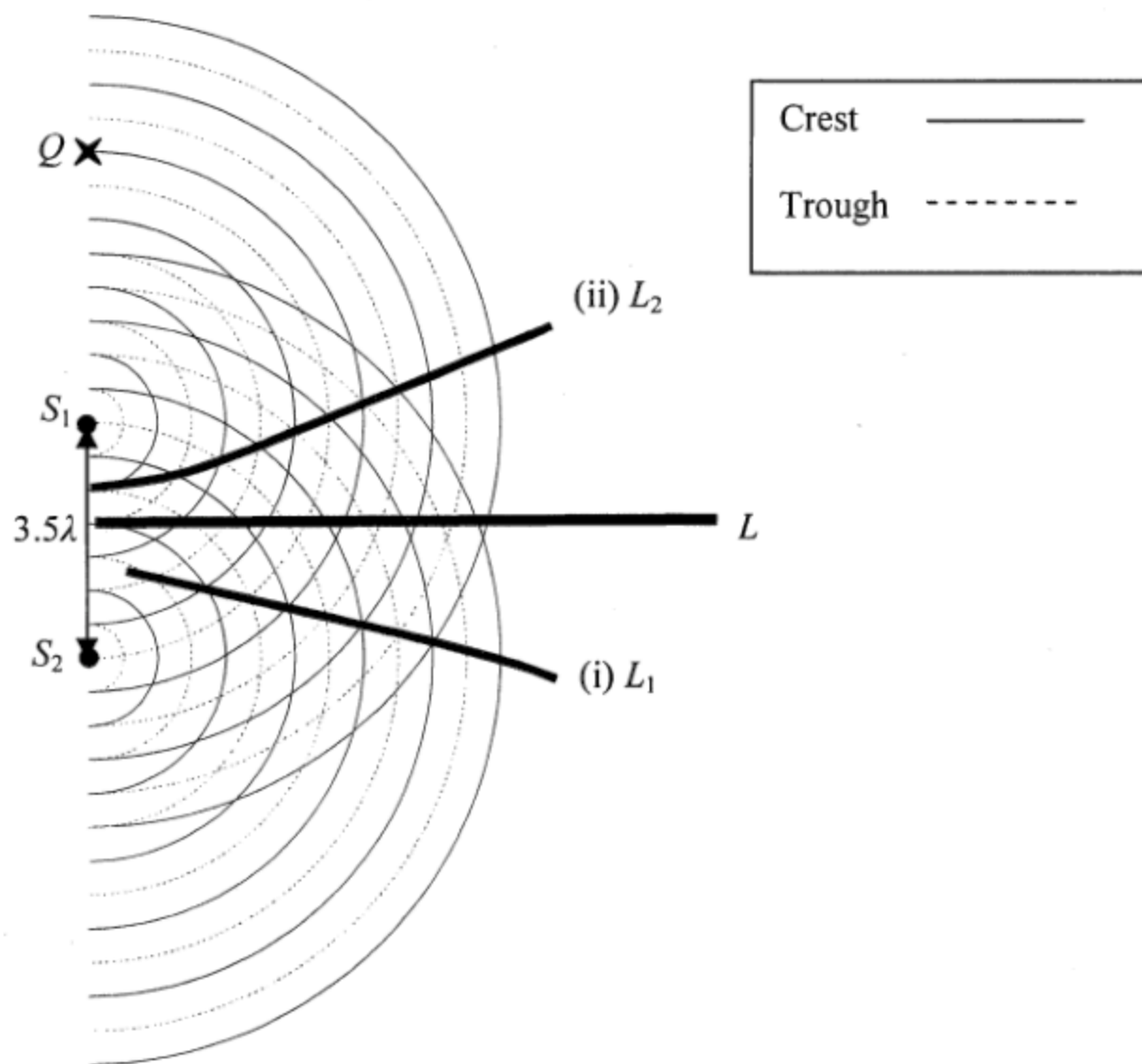
1A

1A

2

5.	(a)	(i)	Let T be the tension. $2T \cos 75^\circ = 60$ $T = 115.9 \text{ N}$	1M	
				1A	<u>2</u>
		(ii)	energy stored in the string = k.e. of arrow $= \frac{1}{2}(0.2)(45)^2$ $= 202.5 \text{ J}$	1M	
				1A	<u>2</u>
	(b)	(i)	$d = v \cos 20^\circ t$ $60 = 45 \cos 20^\circ t$ $t = 1.42 \text{ s}$	1M	
				1A	<u>2</u>
		(ii)	$h = 25 - \frac{1}{2}gt^2$ $= 25 - \frac{1}{2}(9.81)(1.42)^2$ $= 15.1 \text{ m [or } h = 14.9 \text{ m]}$	1M	
				1A	<u>2</u>

6. (a)



2A

L_1 / L_2 are further from L or separation of $L / L_1 / L_2$ is greater or angle between $L / L_1 / L_2$ larger. 1A

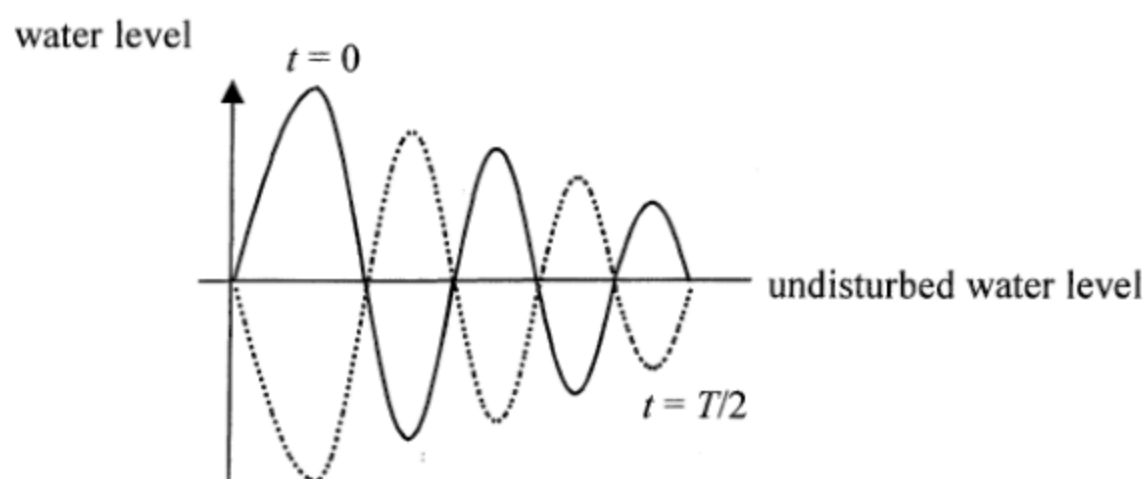
3

60

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Marks

6. (b)



1A

1

(c) The two waves are out of phase at Q as path difference = 3.5λ ($QS_1 = 4\lambda$ and $QS_2 = 7.5\lambda$), destructive interference occurs.

1M

1A

2

(d)

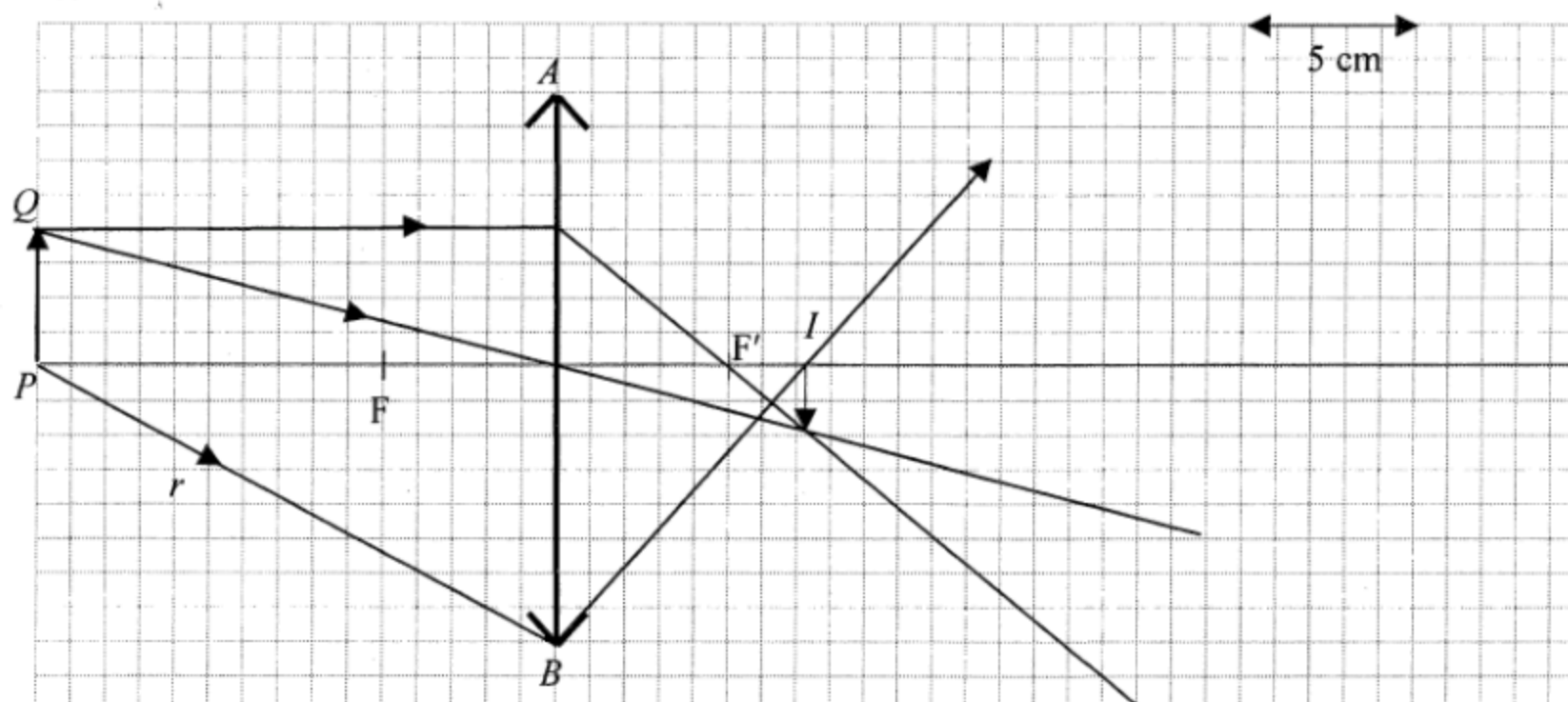
$$\Delta y = \frac{D\lambda}{a} = \frac{2.5 \times 550 \times 10^{-9}}{0.5 \times 10^{-3}} = 2.75 \times 10^{-3} \text{ m}$$

1M

1A

2

7. (a) (i)



Two correct rays to find image I .
Nature: real, inverted, diminished.

2A

2A 4

(ii) Ray r correctly completed.

1A

1

(b) (i) $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ $\frac{1}{15} + \frac{1}{v} = \frac{1}{10}$

1M

$v = 30 \text{ cm}$

1A

$m = \frac{30}{15} = 2$

1A

3

(ii) As the light energy collected by the lens is the same for both cases,
for (b)(i), an enlarged image ($u < v$),
same amount of light energy distributed over a larger
image/intensity of light decreases as distance increases,
i.e. dimmer image for (b)(i)

1A

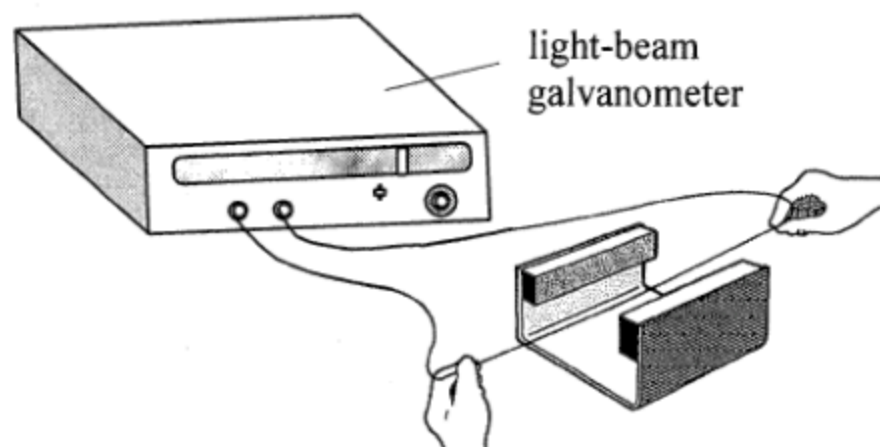
1A

Or for (a), a diminished image ($u > v$),
same amount of light energy distributed over a diminished image,
i.e. brighter image for (a)

2

8. (a)	Keeping warm / 88 W	1A	<u>1</u>
(b)	$R_1 = \frac{V^2}{P} = \frac{220^2}{88}$ $= 550 \Omega$	1M	
		1A	<u>2</u>
(c)	Total current $I_0 = \frac{P_0}{V} = \frac{550}{220} = 2.5 \text{ A}$ Current in R_1, $I_1 = \frac{220}{550} = 0.4 \text{ A}$ Current in R_2, $I_2 = 2.5 - 0.4 = 2.1 \text{ A}$	<u>Or</u> Power to R_2, $550 \text{ W} - 88 \text{ W} = 462 \text{ W}$ Current in R_2, $I_2 = \frac{P_2}{V} = \frac{462}{220}$ $= 2.1 \text{ A}$	1M
		1M	
		1A	<u>3</u>
(d)	Peak current $= \sqrt{2} (2.5 \text{ A})$ $= 3.54 \text{ A}$	1M	
		1A	<u>2</u>

9.	(a)	Humid conditions in bathrooms and water is a conductor which provides a conducting path or lowers the resistance between our hands/body and the source of electricity.	1A 1A	<u>2</u>
	(b)	(i) The person will get electric shock because full potential drop 220 V is applied to or substantial/large current flows through the human body.	1A 1A	<u>2</u>
		(ii) The person will not get electric shock / Nothing happens because the current in secondary circuit has no return path/incomplete circuit.	1A 1A	<u>2</u>
	(c)	Primary : Secondary = 2:1 for 110 V	1A	<u>1</u>



Connect the long wire to the galvanometer./Connect the apparatus as shown.
Put/Place/Move the wire across the magnetic field between the pair of magnets.

Relative movement: Move the wire vertically down across the field and then up across the field, the (light-spot of the) galvanometer would deflect to one side and then to the opposite side.

Polarities of magnet: Move the wire vertically down across the field, the (light-spot of the) galvanometer would deflect to one side and then reverse the polarities of the magnet to repeat the experiment, the galvanometer would deflect to the opposite side.

Direction of movement: Move the wire vertically up/down across the field, the (light-spot of the) galvanometer would have large deflection. No deflection when the wire is moved horizontally to the left/right in the field instead.

Orientation of conductor: With the wire perpendicular to the magnetic field, move it vertically up/down across the field, the (light-spot of the) galvanometer would have large deflection. No deflection when the wire is placed parallel to the field instead.

Rate of movement: Move the wire slowly across the field and then quickly across the field, the (light-spot of the) galvanometer would deflect more for a quicker movement or a faster rate.

No. of turns: Wind the wire into, say, a 10-turn coil and move it again across the field, the (light-spot of the) galvanometer would deflect more for more turns of wire.

ANY
TWO
@2A

11. (a) $^{226}_{88}\text{Ra} \rightarrow ^{222}_{86}\text{Rn} + ^4_2\text{He} + \alpha$ 2A 2
- (b) $\Delta m = 226.0254 - (222.0176 + 4.0026) = 0.0052 \text{ u}$ 1M
 Energy released = $(0.0052)(931) = 4.84 \text{ (MeV)}$ 1A 2
- (c) Number of radium atoms in the source
 $N = N_A \left(\frac{1}{226} \right) \times (5 \times 10^{-6}) = (6.02 \times 10^{23}) \frac{1}{226} \times (5 \times 10^{-6}) = 1.33 \times 10^{16}$ 1A
- Activity $A = \frac{\ln 2}{t_{1/2}} \cdot N$ 1M
 $= \frac{\ln 2}{1600 \times 365 \times 24 \times 3600} \cdot 1.33 \times 10^{16}$
 $= 1.83 \times 10^3 \text{ (disintegrations per second, Bq)}$ 1A 3

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Paper 2

Section A : Astronomy and Space Science

1. D (31%)	2. A (35%)	3. A (53%)	4. C (44%)
5. B (43%)	6. C (36%)	7. D (44%)	8. B (59%)

Marks

1. (a) (i) $L_S = \sigma T_S^4 (4\pi R_S^2)$ 1M
 $L = \sigma T^4 (4\pi R^2)$
 $\therefore \frac{L_S}{L} = \frac{T_S^4 R_S^2}{T^4 R^2}$ 1M
 $R = \left(\frac{T_S}{T} \right)^2 \left(\frac{L}{L_S} \right)^{\frac{1}{2}} R_S$ 2
- (ii)
- | | | | |
|------------|------------------------|------------------|-------|
| Sun | $T_S = 5780 \text{ K}$ | L_S | R_S |
| Betelgeuse | $T = 3650 \text{ K}$ | $L = 126000 L_S$ | R |
- $$R = \left(\frac{5780}{3650} \right)^2 \left(\frac{126000 L_S}{L_S} \right)^{\frac{1}{2}} R_S$$
- $$= 890 R_S$$
- 1M
-
- 1A 2
- (b) (i) $\therefore$ Same brightness measured and brightness $= \frac{L}{4\pi d^2}$ or $\propto \frac{L}{d^2}$ or L increases with d ,
 d is larger $\rightarrow L$ is larger. (for d taken as $(197+45) \text{ pc}$) 1M
 i.e. for Betelgeuse, $\therefore L \propto R^2$ $\therefore R$ increases. 1A 2
- Or
- $$M = m - 5(\log_{10} d - 1) \quad \text{or} \quad M = m - 5(\log_{10} \frac{d}{10})$$

(M : absolute magnitude, m : apparent magnitude)
- (ii) The parallax measurement ($d = \frac{1}{p}$, accurate to within $\sim 100 \text{ pc}$) is too small or d is too large (too far away), ($\sim (1/200)'' = 5 \text{ milliarcsec}$). 1A 1
- (c) $L = 10^9 L_S$, brightness $= \frac{(0.01 \times 10^9 L_S)}{4\pi d^2}$ 1M
 $d = 200 \times 206265 \text{ AU} = 41253000 \text{ AU}$ 1M
 Brightness $= \frac{(0.01 \times 10^9)}{41253000^2} \frac{L_S}{4\pi (1\text{AU})^2} = \frac{(0.01 \times 10^9)}{41253000^2}$ brightness of the Sun 1A
 $= 5.88 \times 10^{-9}$ brightness of the Sun 3

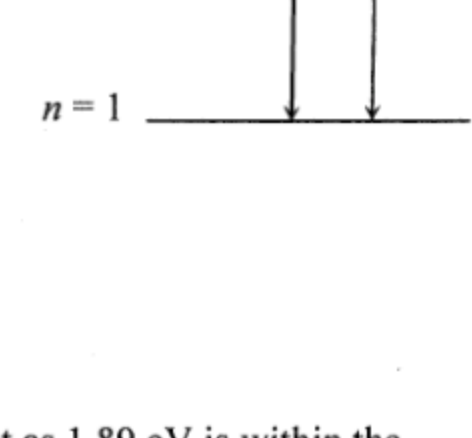
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Section B : Atomic World

1. D (34%)	2. A (47%)	3. B (51%)	4. C (64%)
5. C (53%)	6. A (75%)	7. B (60%)	8. D (47%)

Marks

2. (a) The negative sign means that:
 - the electron is "bound" to the atom
 - the force between the nucleus and electron is attractive
 - work has to be done to remove the electron to infinity } ANY ONE 1A 1
- (b) - The angular momenta of the electrons are whole number multiples of $\frac{h}{2\pi}$,
 i.e. quantized.
 - The electrons are in stable orbits without emitting radiation.
 - The electrons can occupy certain discrete orbits only. / Total energy of the atom is quantized / Energy levels are discrete.
 - The atom can only emit or absorb radiation in the form of a photon when an electron jumps from one energy level to another. } ANY TWO 2A 2
- (c) (i) Energy $= hf$
 $= \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{102.8 \times 10^{-9}} = 1.93 \times 10^{-18} \text{ J}$
 $= 1.93 \times 10^{-18} / (1.60 \times 10^{-19}) = 12.09 \text{ (eV)}$ 1A
 $\Delta E = 12.09 \text{ eV} = - \left(\frac{1}{n^2} - \frac{1}{1^2} \right) 13.6 \text{ eV}$ 1M
 $n^2 = \frac{1}{1 - \frac{12.09}{13.6}}$
 $n^2 = 9.007 \Rightarrow n = 3$ 1A 3
- (ii) The energy of UV = 100.0 nm does not match the differences between ground state of hydrogen and other energy levels. 1A 1
- (iii) There are three possible transitions
 The hydrogen atom is in excited state $n = 3$.
 $E_3 = -\frac{13.6}{3^2} \text{ eV} = -1.51 \text{ eV}$
 $E_2 = -\frac{13.6}{2^2} \text{ eV} = -3.40 \text{ eV}$
 $E_1 = -\frac{13.6}{1^2} \text{ eV} = -13.60 \text{ eV}$
 $3 \rightarrow 1 \quad \Delta E = 12.09 \text{ eV} (= -1.51 - (-13.6))$
 $3 \rightarrow 2 \quad \Delta E = 1.89 \text{ eV} (= -1.51 - (-3.40))$
 $2 \rightarrow 1 \quad \Delta E = 10.2 \text{ eV} (= -3.40 - (-13.6))$
 1M
 The transition from $3 \rightarrow 2$ corresponds to visible light as 1.89 eV is within the corresponding range. 1A 3

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Section C : Energy and Use of Energy

1. A (80%)	2. C (54%)	3. D (43%)	4. C (64%)
5. B (82%)	6. B (68%)	7. C (76%)	8. A (26%)

Marks

3. (a) Maximum solar power per unit area reaching the Earth
 $= 1366 \times (1 - 0.268)$
 $= 1000 \text{ W m}^{-2}$ or W 1A 1
- (b) Solar energy / radiation / radiant energy / light energy $\rightarrow$ electrical energy.
 Adhere a transparent anti-reflection film to the solar cell panel.
 Methods involving tracking the sun to receive maximum sun light
 Mirrors/lens to reflect/collect sun light to the solar cells } ANY ONE 1A 2
- (c) (i) Solar power received by each solar cell
 $= 1000 \text{ W m}^{-2} \times 0.0172 \text{ m}^2$
 $= 17.2 \text{ W}$ 1M
 Electrical power delivered by each solar cell
 $= 17.2 \text{ W} \times 0.12$
 $= 2.064 \text{ W}$ 1M
 Number of solar cells required
 $= \frac{7.35 \text{ kW} \times 4}{2.064 \text{ W}}$
 $= 14244$ 1A 3
- (ii) - To limit/minimize the weight of the aircraft.
 - Limited area for installing the solar cells.
 - It is the batteries to deliver maximum power to the engines, the solar cells are for charging the batteries. } ANY ONE 1A 1
- (d) Energy comes from natural resources / processes that are replenished constantly. 1A
 Wind power 1A
 as (northeast and southwest) monsoons prevail in Hong Kong (during winter and summer). 1A 3

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Section D : Medical Physics

1. A (46%)	2. C (46%)	3. B (55%)	4. B (17%)
5. C (56%)	6. D (52%)	7. D (46%)	8. A (18%)

Marks

4. (a) (i) Let v be the speed of ultrasound in soft tissue
 v_b be the speed of ultrasound in bone
 t be the time of travel of ultrasound in soft tissue
 t_b be the time of travel of ultrasound in bone
 We have $\frac{v_b t_b / 2}{vt / 2} = \frac{5.8}{2.0}$ or $\frac{v_b t_b}{vt} = \frac{5.8}{2.0}$ 1M
 $\left(\frac{v_b}{v} \right) \left(\frac{3}{2} \right) = 2.9$
 $\frac{v_b}{v} = 1.93$ 1A 2
- (ii) From (i), $v_b = 1.93 \times 1580 \text{ m s}^{-1} = 3055 \text{ m s}^{-1}$ 1M
- $\therefore$ for bone $Z = \rho c$
 $7.78 \times 10^6 = \rho (3055)$
 $\rho = 2547 \text{ kg m}^{-3}$ 1M
 1A 3
- (b) (i) Reflection of ultrasound at tissue boundary/when entering another tissue. 1A
 The brightness/amplitude/strength in a B-scan image is proportional to the intensity of the reflected ultrasound signal/change in acoustic impedance. 1A
 Distance/depth calculated from time for signal to return to ultrasound transducer./When traces are joined, a 2D/planar image of the scanned area is formed. 1A 3
- (ii) Advantages (ANY ONE): 1A
 1. relatively safe (as no ionizing radiation involved)
 2. readily accessible
 3. can detect the movement of an organ in real time
- Limitations (ANY ONE): 1A
 1. limited tissue penetration, particularly through bones or air-filled structures
 2. field-of-view (FOV) of ultrasound is smaller compared to other methods of imaging 2

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